

- 17 Show that $\frac{2\sqrt{7}+2}{\sqrt{7}-3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

$$\begin{aligned} & \frac{2\sqrt{7}+2}{\sqrt{7}-3} \times \frac{(\sqrt{7}+3)}{(\sqrt{7}+3)} \\ & \frac{2\sqrt{7}(\sqrt{7}+3) + 2(\sqrt{7}+3)}{7-9} \\ & \frac{2 \times 7 + 6\sqrt{7} + 2\sqrt{7} + 6}{-2} \\ & \frac{14 + 6 + 6\sqrt{7} + 2\sqrt{7}}{-2} \\ & = \frac{20 + 8\sqrt{7}}{-2} \end{aligned}$$

$$\begin{aligned} & \frac{20 + 8\sqrt{7}}{-2} \\ & -2(-10 - 4\sqrt{7}) \end{aligned}$$

(Total for Question 17 is 3 marks)

- 18 Make y the subject of $p = \sqrt{\frac{y+w}{3y-t}}$

$$p = \sqrt{\frac{y+w}{3y-t}}$$

$$3y-t \times p^2 = \frac{y+w}{3y-t} \times 3y-t$$

$$3yp^2 + tp^2 = y + w$$

$$3yp^2 - y = w - tp^2$$

$$y(3p^2 - 1) = \frac{w - tp^2}{3p^2 - 1}$$

$$y = \frac{w - tp^2}{3p^2 - 1}$$

$$y = \frac{w - tp^2}{3p^2 - 1}$$

(Total for Question 18 is 4 marks)

DO NOT WRITE IN THIS AREA

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